

Time Series Momentum Strategy with Volatility Targeting

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Aug 2025

Abstract

This project implements a simplified time-series momentum (TSMOM) strategy across a basket of exchange-traded funds (ETFs). Momentum signals are constructed using multiple lookback horizons and aggregated into a directional trading signal, while positions are scaled using volatility targeting to control risk exposure. The strategy is evaluated using a rolling walk-forward backtesting framework, with a 3-year training window and 1-year out-of-sample testing period. Performance is assessed using standard risk-adjusted metrics and compared against a benchmark strategy.

1 Introduction

Time-series momentum (TSMOM) refers to the tendency of asset returns to exhibit persistence over time, where past returns can help predict future returns. This phenomenon has been widely documented across asset classes and forms the basis of many systematic trading strategies, particularly in global macro and managed futures contexts.

The objective of this project is to:

- Implement a simplified TSMOM strategy using a diversified set of ETFs.
- Evaluate its performance under a backtesting framework.
- Compare performance metrics against those of a benchmark strategy.
- Analyse its strengths and limitations.

The project will focus on simplicity and interpretability rather than maximizing performance.

2 Data

The dataset consists of historical daily price data for a selection of ETFs representing different asset classes.

2.1 Chosen ETFs

To ensure diversification, a mixture of US and non-US assets were chosen:

- AGG: US Bonds

- DBC: US Commodities
- EEM: Emerging Markets
- EFA: Developed markets ex US
- GLD: Gold
- SPY: US Equities
- VNQ: US Real Estate

2.2 Data Processing

- Prices were aligned to a common business-day calendar.
- Missing values were forward-filled to eliminate gaps.

The resulting dataset was subsequently used as input for signal generation and backtesting.

3 Methodology

As an overview, the strategy consists of the following main components:

1. Signal Generation
2. Position Sizing via Volatility Targeting
3. Rolling Walk Forward backtesting

3.1 Signal Construction

Momentum signals were computed in a similar fashion as described in Hurst et al. (2017) which uses multiple lookback horizons over 1 month (63 days), 3 months (126 days), and 6 months (252 days).

For each horizon, cumulative past returns were computed. A signal of +1 was assigned to positive returns, and -1 to negative returns.

Then, signals across horizons were averaged to produce a smoother and more robust final signal.

3.2 Volatility Targeting

Volatility targeting was used to stabilise risk exposure across assets. Without it, assets exhibit heterogeneous volatility and high-volatility ones would likely dominate portfolio risk.

3.2.1 Volatility Weighing

1. Annualized volatility was computed from daily percentage changes.
2. Positions were scaled in a manner inversely proportionate to their volatilities as described in Moskowitz et al. (2012). For example, one with a higher volatility would take on a smaller position size.

3. Weekly and then daily resamples were computed before being further normalised to ensure a sum of 1, accounting for ETFs with missing historical data.

3.2.2 En-Ante Volatility Targeting

1. Positions were scaled to achieve 10% annualized portfolio volatility as with the procedure described in Hurst et al. (2017).
2. Computations were simplified by considering only each asset’s individual variance – this utilised diagonal approximation to ignore the covariance between the assets themselves.
3. Weights were shifted forwarded by 1 day to avoid look-ahead bias.

3.3 Rolling Walk Forward Analysis

- Data was partitioned into overlapping windows of 3-years of training data and 1 year of out-of-sample (OOS) testing data that was continuously applied on a rolling basis (forwarded 1 year for every iteration).
- A fixed number of days based on the maximum and minimum lookback period was used as a padding before each test window to ensure sufficient historical data for calculations. Subsequently, each test window filtered out ETFs with insufficient historical data.
- OOS returns were concatenated.

3.4 Benchmark Strategy

An equal-weighted buy-and-hold portfolio of the same basket of ETFs was chosen as the benchmark strategy. Our goal was to determine whether a naive holding of the same basket would perform better or worse than the proposed TSMOM strategy.

3.5 Metrics and Plotting

- Cumulative equity curves (growth of 1\$) and drawdowns of both the TSMOM and benchmark strategies were plotted.
- Calculated metrics for both strategies include
 - Annualised return
 - Annualised volatility
 - Sharpe ratio
 - Maximum drawdown

over the same period.

4 Interpretation of Results and Discussion

4.1 Results

The plots comparing the performance of the strategies are shown below:

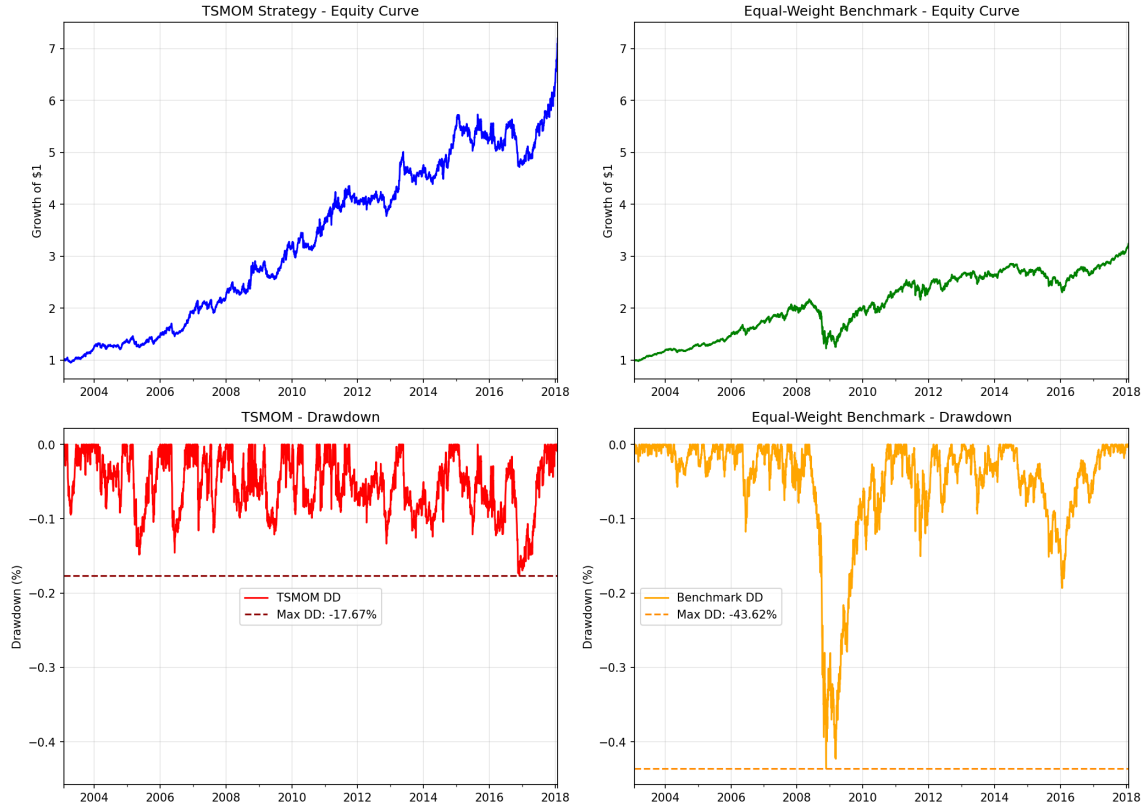


Figure 1: Comparison of equity curve and drawdown plots of TSMOM and benchmark strategies

Metric	TSMOM	Benchmark
Annualised Return	13.82%	8.64%
Sharpe Ratio	0.9274	0.5928
Maximum Drawdown	17.67%	43.62%

Table 1: Key metrics of TSMOM and benchmark strategies

4.2 Interpretation

TSMOM generates significantly higher returns and a lower maximum drawdown than simply holding equal weights of the same ETFs over the same period. This can be attributed to the strategy's inherent nature to capture persistent price trends while stabilising risk through volatility targeting.

4.3 Limitations

- TSMOM was only tested on a fixed basket of ETFs over a singular duration. Without further testing on other ETFs and other securities, particularly derivatives, and throughout further durations, we are unable to make a definite conclusion on TSMOM’s overall effectiveness.
- Adding on to the previous point, the market regimes during the duration in which TSMOM was tested were not analysed. Therefore, TSMOM could ultimately underperform in a different set of regimes.
- The chosen basket of ETFs were introduced at varying points in time with most starting only from the early 2000s. The data is thus dependent on the effectiveness of weight normalisations.
- The volatility lookback period was arbitrarily chosen to be of three months without strong rationale. This may be insufficient, or even too long, of a period.
- Diagonal approximation was assumed to hold when computing volatility targeting so that calculations could be simplified. This may have resulted in an under or overestimation of the true portfolio risk.
- Transaction costs were not accounted for.

5 Conclusion

This project implemented a simplified time-series momentum strategy with volatility targeting, with evaluation performed through a rolling backtesting framework. When compared against a benchmark strategy, TSMOM’s metrics prevail significantly.

The implemented TSMOM strategy is however subject to many limitations, particularly the lack of testing on other securities and market regimes. Despite its strong performance relative to a naive hold of equally weighted assets, it may ultimately underperform for different securities or in a different regime.

References

- Hurst, B., Ooi, Y. H., and Pedersen, L. H. (2017). A century of evidence on trend-following investing. *Journal of Portfolio Management*, 44(1):15–29.
- Moskowitz, T. J., Ooi, Y. H., and Pedersen, L. H. (2012). Time series momentum. *Journal of Financial Economics*, 104(2):228–250.